TEST 2

ANSWERS

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BINARY

Question one

1. An overflow counts for about 23% according to the CVSS of the attack and therefore, requires deep knowledge and understanding in order to easily identify and mitigate it as a vulnerability. This introduces one's and two's compliment numbers in binary number system which the computer is believed to be using.

a) Add the following numbers in two's compliment using 8 bits. Determine whether an overflow has occurred. Show all your working. [6 Marks]

BINARY (CONTINUES…)

i. -47 and 19

-47 + 19 or 19 – 47 = -28

19 + (-47)

	0	0	0	1	0 0 1 1	= 19
+	1	1	0	1	0 0 0 1	= 47's 2's comp.
	1	1	1	0	0 1 0 0	

128	64	32	16	8	4	2	1	
0	0	0	1	0	0	1	1	= 19
0	0	1	0	1	1	1	1	= 47
1	1	0	1	0	0	0	0	= 47's 1's compliment
+							1	
1	1	0	1	0	0	0	1	= 47's 2's compliment

BINARY (CONTINUES…)

ii. 98 and 74

98 + 74 = 172

128	64	32	16	8	4	2	1	
0	1	1	0	0	0	1	0	= 98
+ 0	1	0	0	1	0	1	0	= 74
1	0	1	0	1	1	0	0	= 172

An overflow has occurred

BINARY (CONTINUES…)

b) Add the following numbers in one's compliment using 8 bits. Determine whether an overflow has occurred. [6 Marks]

i. 25 and -56

25 + (-56) = -31

	0	0	0	1	1	0	0	1 = 25
+	1	1	0	0	0	1	1	1 = 56's 1's comp.
	1	1	1	0	0	0	0	0 = -31

128	64	32	16	8	4	2	1	
0	0	0	1	1	0	0	1	= 25
0	0	1	1	1	0	0	0	= 56
1	1	0	0	0	1	1	1	= 56's 1's comp.

BINARY (CONTINUES…)

ii. -77 and -77

-77 + (-77) = -154

	1 0 1	1 0 0 1 0 = 77's 1's comp.
+	1 0 1	1 0 0 1 0 = 77's 1's comp.
	0 1 1	0 0 1 0 0
+		1
	0 1 1	0 0 1 0 1

An overflow as occurred

128	64 32 16 8 4 2 1	
0	1 0	0 1 1 0 1 = 77
1	0 1	1 0 0 1 0 = 77's 1's compliment

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BINARY (CONTINUES…)

c) Subtract the following numbers in one's compliment using 8 bits. Determine whether an overflow has occurred.

i. 66 – (-78) = 144

This is the same as 66 + (78)

128	64	32	16	8	4	2	1	
0	1	0	0	0	0	1	0	= 66
+ 0	1	0	0	1	1	1	0	= 78
1	0	0	1	0	0	0	0	

An overflow as occurred

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BINARY (CONTINUES…)

ii. 72 – 25 = 47 is 72 + (-25)

	0	1	0	0	1	0	0	0 = 72
+	1	1	1	0	0	1	1	0 = 25's 1's Compliment
	0	0	1	0	1	1	1	0
+								1
	0	0	1	0	1	1	1	1

128	64	32	16	8	4	2	1	
0	1	0	0	1	0	0	0	= 72
0	0	0	1	1	0	0	1	= 25
1	1	1	0	0	1	1	0	= 25's 1's Compliment

BINARY (CONTINUES…)

d) Subtract the following numbers in two's compliment using an octet. Determine whether an overflow has occurred. [6 Marks]

i. 70 – 32

70 + (-32) = 38

	0	1	0	0	0	1	1	0 = 70
+	1	1	1	0	0	0	0	0 = 32's 2's comp.
	0	0	1	0	0	1	1	0 = 38

128	64	32	16	8	4	2	1	
0	1	0	0	0	1	1	0	= 70
0	0	1	0	0	0	0	0	= 32
1	1	0	1	1	1	1	1	= 32's 1's comp.
+							1	
1	1	1	0	0	0	0	0	= 32's 2's comp.

BINARY (CONTINUES…)

ii. 99 – (71)

99 + (-71) = 28

0	1 1	0 0 0 1 1 = 99
+ 1	0 1	1 1 0 0 1 = 71's 2's comp.
0	0 0	1 1 1 0 0 = 28

128	64 32	16 8 4 2 1
0	1 1	0 0 0 1 1 = 99
0	1 0	0 0 1 1 1 = 71
1	0 1	1 1 0 0 0 = 71's 1's comp.
+		1
1	0 1	1 1 0 0 1 = 71's 2's comp.

BINARY (CONTINUES…)

2. What is an overflow? [1 Mark]

   Based on these calculations, an overflow occurs when you expect your result as a positive or negative and get a different result. For example, adding two positive numbers and getting a negative result.

PERT CHART

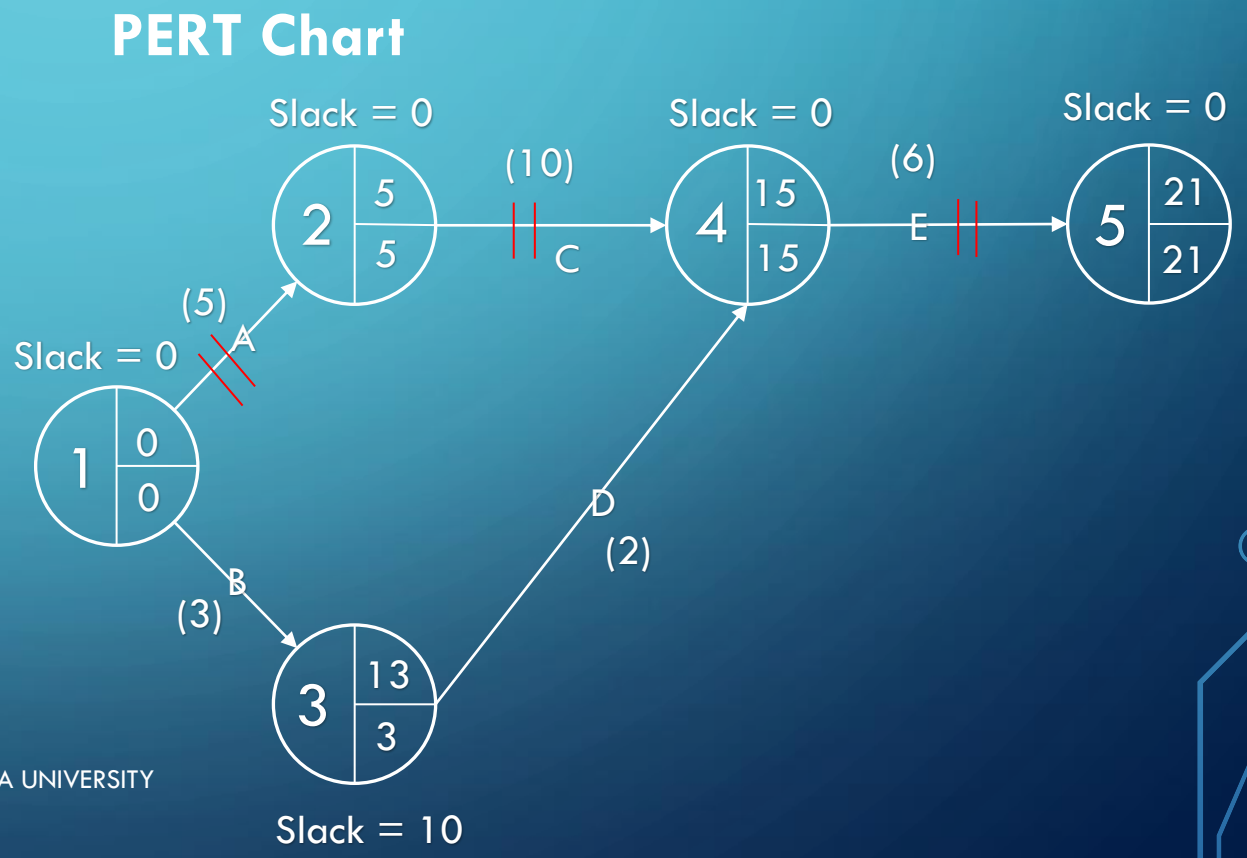
Question two [25 Marks]

1. Mitigation and Remediation require planning in advance. PERT can be used to visualize these plans and make it easy for IT and Project Managers to follow through.

a) Construct a network diagram with the data given below. [5 Marks]

PERT CHART (CONTINUES…)

Activities	Dependencies	Duration (Days)
A		5
B		3
C	A	10
D	B	2
E	C,D	6



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PERT CHART (CONTINUES…)

b) Compute the Project duration. [2 Marks]

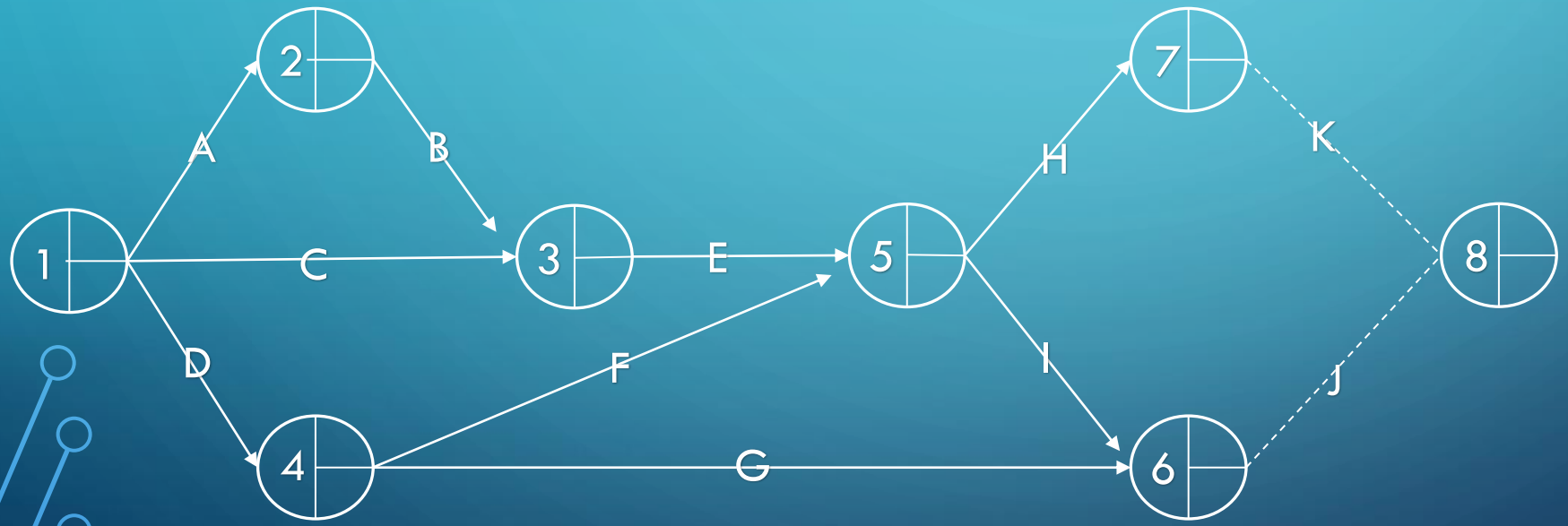
Duration of The Project = 5 + 10 + 6
= 21 Days

c) Determine the critical path. [10 Marks]

Critical Path (CMP) = 1 – 2 – 4 – 5

d) Draw a project network for a remediation plan from the following statement E > B, C; F, G > D; H,I > E, F; J > I, G; K > H; B > A;. [8 Marks]

PERT CHART (CONTINUES…)



Activities	Predecessors
A	
B	A
C	
D	
E	B, C
F	D
G	D
H	E, F
I	E, F
J	I, G
K	H

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PERT CHART (CONTINUES…)

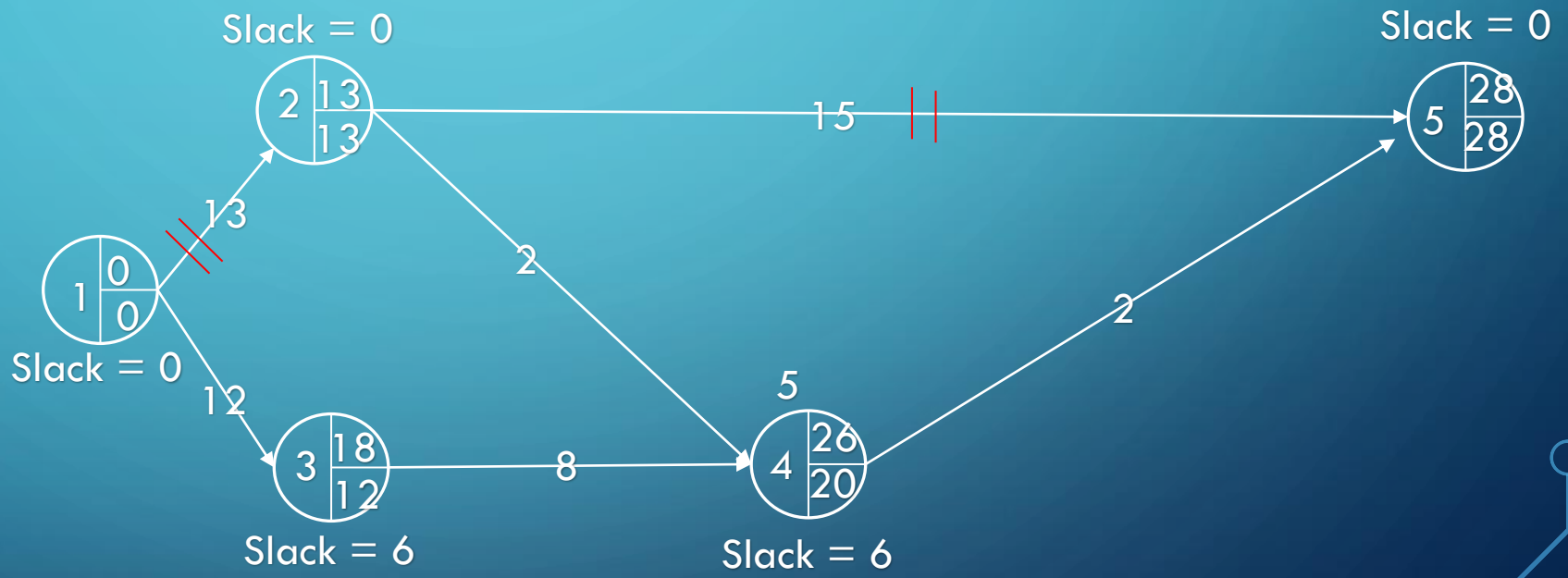
Question Three [25 Marks]

1. Activities of a remediation plan has been assigned to you to come up with the project network which is scheduled to be presented in a managing meeting.

a) Draw a network diagram for the remediation plan project. [5 Marks]

PERT CHART (CONTINUES…)

Activities	Duration
1 – 2	13
1 – 3	12
2 – 4	2
3 – 4	8
2 – 5	15
4 – 5	2



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PERT CHART (CONTINUES…)

b) Determine the critical path and find the Earliest Start (ES), Earliest Finish (EF), Latest Start (LS), and Latest Finish (LF) Time. [10 Marks]

Critical Path = 1 – 2 – 5

PERT CHART (CONTINUES…)

Activities	ES	Duration	EF (ES + D)
1 – 2	0	13	13
1 – 3	0	12	12
2 – 4	13	2	15
3 – 4	12	8	20
2 – 5	13	15	28
4 – 5	20	2	22

Activities	LF	Duration	LS (LF - D)
1 – 2	13	13	0
1 – 3	18	12	6
2 – 4	26	2	24
3 – 4	26	8	18
2 – 5	28	15	13
4 – 5	28	2	26

LS	ES	TF (LS - ES)
0	0	0
6	0	6
24	13	11
18	12	6
13	13	0
26	20	6

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PERT CHART (CONTINUES…)

d) Calculate the Total Float (TF), Free Float (FF), Independent Float (IND F), and the Interfering Float (INT F). [10 Marks]

PERT CHART (CONTINUES…)

Activities	T	H	TF	FF (TF – H)	IND F (FF – T)	INT F (TF – FF)
1 – 2	0	0	0	0	0	0
1 – 3	0	6	6	0	0	6
2 – 4	0	6	11	5	5	6
3 – 4	6	6	6	0	-6	6
2 – 5	0	0	0	0	0	0
4 – 5	6	0	6	6	0	0

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PERT CHART (CONTINUES…)

Question four [25 Marks]

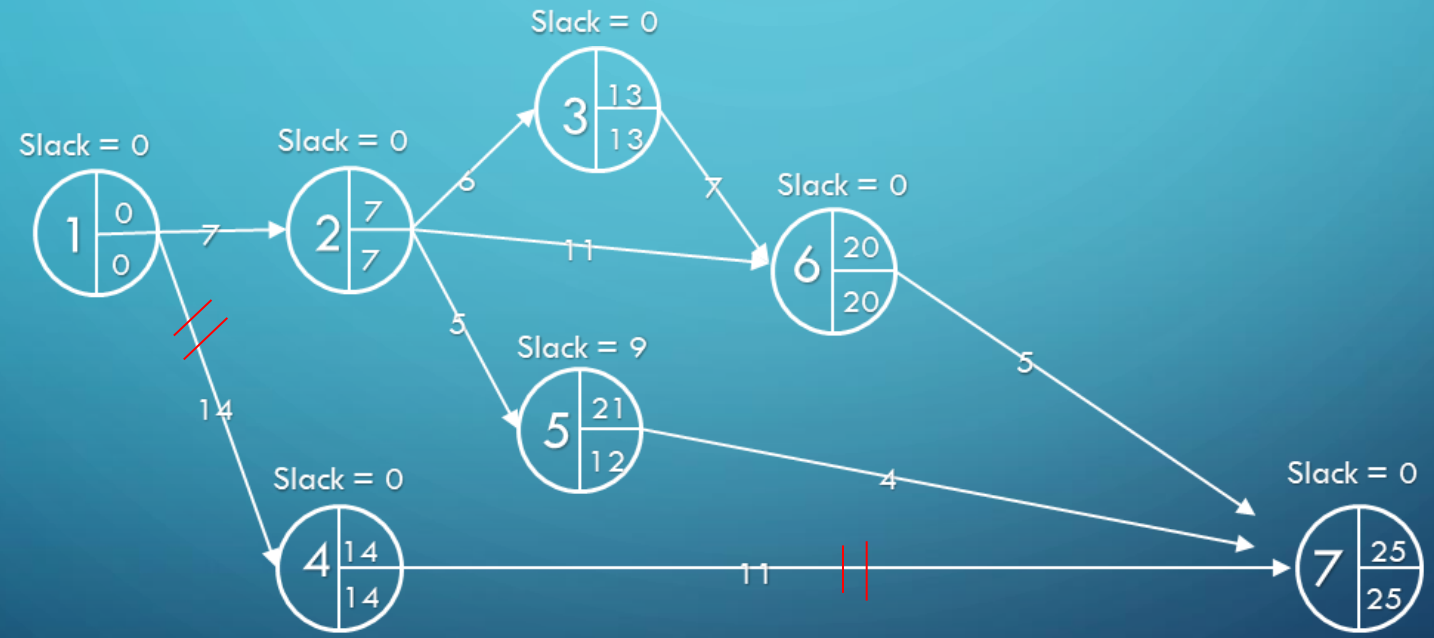
The following have been given for your convenient as an IT Manager in charge of all ICT work related project.

Activity	Optimistic (O or to)	Pessimistic (P or tp)	Most Likely (M or tm)
1 – 2	3	15	6
2 – 3	2	14	5
1 – 4	6	30	12
2 – 5	2	8	5
2 – 6	5	17	11
3 – 6	3	15	6
4 – 7	3	27	9
5 – 7	1	7	4
6 – 7	2	8	5

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PERT CHART (CONTINUES…)

a) Draw the corresponding network diagram.



PERT CHART (CONTINUES…)

b) Compute the expected duration for each activity. [5 Marks]

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4tm	Te = to + 4tm + tp
4(6) = 24	(3 + 24 + 15) / 6 = 7
4(5) = 20	(2 + 20 + 14) / 6 = 6
4(12) = 48	(6 + 48 + 30) / 6 = 14
4(5) = 20	(2 + 20 + 8) / 6 = 5
4(11) = 44	(5 + 44 + 17) / 6 = 11
4(6) = 24	(3 + 24 + 15) / 6 = 7
4(9) = 36	(3 + 36 + 27) / 6 = 11
4(4) = 16	(1 + 16 + 7) / 6 = 4
4(5) = 20	(2 + 20 + 8) / 6 = 5

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PERT CHART (CONTINUES…)

c) Calculate the expected variance of each activity. [5 Marks]

σ^2
4
4
16
1
4
4
16
1
1

PERT CHART (CONTINUES…)

d) Determine the expected critical path variance. [5 Marks]

Critical Path = 1 – 4 – 7

Critical Variance = 16 + 16
= 32

e) Compute the standard deviation of each activity. [5 Marks]

σ = (tp – to) / 6
(15 – 3) / 6 = 2
(14 – 2) / 6 = 2
(30 – 6) / 6 = 4
(8 – 2) / 6 = 1
(17 – 5) / 6 = 2
(15 – 3) / 6 = 2
(27 – 3) / 6 = 4
(7 – 1) / 6 = 1
(8 – 2) / 6 = 1

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